IELTS ACADEMIC READING

Volume 10

3 complete Academic Reading tests — three full passages and 40 questions each — with a complete answer key and passage evidence for every answer

IB

3	120	Full
FULL TESTS	QUESTIONS	ANSWER KEY

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Introduction

Welcome to **IELTS Academic Reading — Volume 10**, a printable practice book of 3 complete Reading tests, compiled and prepared by Teacher Ibodullaev Bekhruz.

Every test follows the real exam format: three passages of increasing difficulty and 40 questions covering the full range of IELTS question types — True/False/Not Given, Matching Headings, Matching Information, Multiple Choice, Summary and Sentence Completion, and more. Answer circles and blank lines are left for you to write your answers, just like on test day.

Part One contains all 3 tests with questions only, so you can practise without seeing any answers. **Part Two** contains the complete Answer Key for each test, with the keywords from each question and the exact sentence in the passage that proves the correct answer.

How to use this book:

- Sit each test in one go, under timed conditions — 60 minutes for all three passages.
- Write your answers directly in the circles and blanks provided.
- Check your answers in Part Two and read the passage evidence for every question you got wrong.
- Use the band-score table on the answer key pages to estimate your result, then revisit weak question types.

Good luck with your preparation, and follow **@bekhruzposts** for more free IELTS materials.

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IELTS Academic Reading — Test 1

Volume 10 · 3 Passages · 40 Questions

Reading Passage 1

The Formation of Continents and the Evolution of Species

A The fact that there was once a Pangean supercontinent, a Panthalassa Ocean, and a Tethys Ocean, has profound implications for the evolution of multicellular life on Earth. These considerations were unknown to the scientists of the 19th century — making their scientific deductions even more remarkable. Quite independently of each other, Charles Darwin and his young contemporary Alfred Russel Wallace reached the conclusion that life had evolved by natural selection. Wallace later wrote in My Life of his own inspiration.

B Why do some species die and some live? The answer was clearly that on the whole the best fit lived. From the effects of disease, the most-healthy escaped; from enemies the strongest, the swiftest or the most cunning; from famine the best hunters. Then, it suddenly flashed on me that this self-acting process would improve the race, because in every generation the inferior would inevitably be killed off and the superior would remain, that is, the fittest would survive.

C Both Darwin's and Wallace's ideas about natural selection had been influenced by the essays of Thomas Malthus in his Principles of Population. Their conclusions, however, had been the direct result of their personal observation of animals and plants in widely separated geographic locations: Darwin from his experiences during the voyage of the Beagle, and particularly during the ship's visit to the Galapagos Islands in the East Pacific in 1835; Wallace during his years of travel in the Amazon Basin and in the Indonesia-Australian Archipelago in the 1850s.

D Darwin had been documenting his ideas on natural selection for many years when he received a paper on this same subject from Wallace, who asked for Darwin's opinion and help in getting it published. In July 1858, Charles Lyell and J. D Hooker, close friends of Darwin, pressed Darwin to present his conclusions so that he would not lose priority to an unknown naturalist. Presiding over the hastily called but now historic meeting of the Linnean Society in London, Lyell and Hooker explained to the distinguished members how "these two gentlemen" (who were absent: Wallace was abroad, and Darwin chose not to attend), had "independently and unknown to one another, conceived the same very ingenious theory."

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E Both Darwin and Wallace had realized that the anomalous distribution of species in particular regions had profound evolutionary significance. Subsequently, Darwin spent the rest of his days in almost total seclusion thinking and writing mainly about the origin of species. In contrast, Wallace applied himself to the science of biogeography, the study of the pattern and distribution of species, and its significance, resulting in the publication of a massive two volume work the Geographical Distribution of Animals in 1876.

F Wallace was a gentle and modest man, but also persistent and quietly courageous. He spent years working in the most arduous possible climates and terrains, particularly in the Malay Archipelago; he made patient and detailed zoological observations and collected huge numbers of specimens for museums and collectors — which is how he made a living. One result of his work was the conclusion that there is a distinct faunal boundary, called "Wallace's line," between an Asian realm of animals in Java, Borneo and the Philippines and an Australian realm in New Guinea and Australia. In essence this boundary posed a difficult question: How on Earth did plants and animals with a clear affinity to the Northern Hemisphere meet with their Southern Hemispheric counterparts along such a distinct Malaysian demarcation zone? Wallace was uncertain about demarcation on one particular island — Celebes, a curiously shaped place that is midway between the two groups. Initially he assigned its flora-fauna to the Australian side of the line, but later he transferred it to the Asian side. Today we know the reason for his dilemma. 200 MYA East and West Celebes were islands with their own natural history lying on opposite sides of the Tethys Ocean. They did not collide until about 15 MYA. The answer to the main question is that Wallace's Line categorizes Laurasia-derived flora-fauna (the Asian) and Gondwana-derived flora-fauna (the Australian), fauna that had evolved on opposing shores of the Tethys. The closure of the Tethys Ocean today is manifested by the ongoing collision of Australia/New Guinea with Indochina/Indonesia and the continuing closure of the Mediterranean Sea — a remnant of the Western Tethys Ocean.

G In his ORIGIN OF CONTINENTS AND OCEANS, Wegener quoted at length from Wallace's Geographical Distribution of Animals. According to Wegener's reading, Wallace had identified three clear divisions of Australian animals, which supported his own theory of continental displacement. Wallace had shown that animals long established in southwestern Australia had an affinity with animals in South Africa, Madagascar, India, and Ceylon, but did not have an affinity with those in Asia. Wallace also showed that Australian marsupials and monotremes are clearly related to those in South America, the Moluccas, and various Pacific islands, and that none are found in neighboring Indonesia. From this and related data, Wegener concluded that the then broadly accepted "landbridge" theory could not account for this distribution of animals and that only his theory of continental drift could explain it.

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H The theory that Wegener dismissed in preference to his own proposed that plants and animals had once migrated across now-submerged intercontinental landbridges. In 1885, one of Europe's leading geologists, Eduard Suess, theorized that as the rigid Earth cools, its upper crust shrinks and wrinkles like the withering skin of an aging apple. He suggested that the planet's seas and oceans now fill the wrinkles between once-contiguous plateaus.

I Today, we know that we live on a dynamic Earth with shifting, colliding and separating tectonic plates, not a "withering skin," and the main debate in the field of biogeography has shifted. The discussion now concerns "dispersalism" versus "vicarianism" — unrestricted radiation of species on the one hand and the development of barriers to migration on the other. Dispersion is a short-term phenomenon — the daily or seasonal migration of species and their radiation to the limits of their natural environment on an extensive and continuous landmass. Vicarian evolution, however, depends upon the separation and isolation of a variety of species within the confines of natural barriers in the form of islands, lakes, or shallow seas — topographical features that take a long time to develop.

Questions 1–5

Use the information in the passage to match the people with opinions or deeds below. Choose the appropriate letters.

LIST OF PEOPLE

A Suess
B Wallace
C Darwin and Wallace
D Wegener
E Lyell and Hooker

Q	Statement	Answer
1	Urged Darwin to publish his scientific findings	
2	Depicted physical features of earth's crust	
3	Believed in continental drift theory while rejecting another one	
4	Published works about wildlife distribution in different regions	
5	Evolution of species is based on selection by nature	

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Questions 6–8

Reading Passage 1 has nine paragraphs, **A–I**. *Which paragraph contains the following information?*

Q	Statement	Answer
6	The most adaptable would survive	
7	The boundary called Wallace's line is found between Asia and Australia	
8	Animal affinity exists between Australia and Africa	

Questions 9–13

Complete the following summary of the paragraphs of Reading Passage. Write **NO MORE THAN TWO WORDS** *from the Reading Passage for each answer.*

Wegener found that continental drift, instead of "land bridge" theory in which vegetation and wildlife (9) ———— intercontinentally, could explain strange species' distribution phenomenon. However, Eduard Suess compared the wrinkle of the crust to the (10) ———— of an old apple. Now it is well known that we are living on the planet where there are (11) ———— in constant mobile states instead of what Suess described. The debate in biogeography has switched to concerns between two terms: "(12) ————" and "(13) ————".

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Reading Passage 2

Demystifying Yawn

A When a scientist began to study yawning in the 1980s, it was difficult to convince some of his research students of the merits of "yawning science." Although it may appear quirky, his decision to study yawning was a logical extension to human beings of his research in developmental neuroscience, reported in such papers as "Wing-flapping during Development and Evolution." As a neurobehavioral problem, there is not much difference between the wing-flapping of birds and the face and body flapping of human yawners.

B Yawning is an ancient, primitive act. Humans do it even before they are born, opening wide in the womb. Some snakes unhinge their jaws to do it. One species of penguins yawns as part of mating. Only now are researchers beginning to understand why we yawn, when we yawn and why we yawn back. A professor of cognitive neuroscience at Drexel University in Philadelphia, Steven Platek, studies the act of contagious yawning, something done only by people and other primates.

C In his first experiment, he used a psychological test to rank people on their empathic feelings. He found that participants who did not score high on compassion did not yawn back. "We literally had people saying, 'Why am I looking at people yawning?'" Professor Platek said. "It just had no effect."

D For his second experiment, he put 10 students in a magnetic resonance imaging machine as they watched video tapes of people yawning. When the students watched the videos, the part of the brain which reacted was the part scientists believe controls empathy — the posterior cingulate, in the brain's middle rear. "I don't know if it's necessarily that nice people yawn more, but I think it's a good indicator of a state of mind," said Professor Platek. "It's also a good indicator if you're empathizing with me and paying attention."

E His third experiment is studying yawning in those with brain disorders, such as autism and schizophrenia, in which victims have difficulty connecting emotionally with others. A psychology professor at the University of Maryland, Robert Provine, is one of the few other researchers into yawning. He found the basic yawn lasts about six seconds and they come in bouts with an interval of about 68 seconds. Men and women yawn or half-yawn equally often, but men are significantly less likely to cover their mouths, which may indicate complex distinction in genders. "A watched yawner never yawns," Professor Provine said. However, the physical root of yawning remains a mystery. Some researchers say it's coordinated within the hypothalamus of the brain, the area that also controls breathing.

F Yawning and stretching also share properties and may be performed together as parts of a global motor complex. But they do not always co-occur; people usually yawn when we stretch, but we don't always stretch when we yawn, especially before bedtime. Studies by

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G.H.A. Visser and H.F. Prechtl in the early 1980s, charting movement in the developing fetus using ultrasound, observed not just yawning but a link between yawning and stretching as early as the end of the first prenatal trimester.

G The most extraordinary demonstration of the yawn-stretch linkage occurs in many people paralyzed on one side of their body because of brain damage caused by a stroke. The prominent British neurologist Sir Francis Walshe noted in 1923 that when these hemiplegics yawn, they are startled and mystified to observe that their otherwise paralyzed arm rises and flexes automatically in what neurologists term an "associated response." Yawning apparently activates undamaged, unconsciously controlled connections between the brain and the cord motor system innervating the paralyzed limb. It is not known whether the associated response is a positive prognosis for recovery, nor whether yawning is therapeutic for reinnervation or prevention of muscular atrophy.

H Clinical neurology offers other surprises. Some patients with "locked-in" syndrome, who are almost totally deprived of the ability to move voluntarily, can yawn normally. The neural circuits for spontaneous yawning must exist in the brain stem near other respiratory and vasomotor centers, because yawning is performed by anencephalic individuals who possess only the medulla oblongata. The multiplicity of stimuli of contagious yawning, by contrast, implicates many higher brain regions.

Questions 14–18

Complete the summary paragraph described below. Write **NO MORE THAN THREE WORDS** *from the Reading Passage for each answer.*

A psychology professor drew a conclusion that an average yawn lasts about 6 seconds followed by an (14) _____ before another yawn. It is almost at the same frequency that males and females yawn or half yawn, but behavior accompanied with yawning shows a (15) _____ in genders. Besides, a part of the brain involved in yawning may also be responsible for (16) _____. Another study found that there is a link between a yawn and (17) _____ during the first three months of pregnancy, and yawning was also shown to set off an involuntary movement in people with (18) _____.

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Questions 19–23

Reading Passage 2 has eight paragraphs, **A–H**. *Which paragraph contains the following information? NB You may use any letter more than once.*

Q	Statement	Answer
19	The rate for yawning shows some regular pattern.	
20	Yawning is an inherent ability that appears in both animals and humans.	
21	Stretching and yawning doesn't always occur together.	
22	Yawning may suggest people are empathetic and attentive.	
23	Some superior areas in the brain may underline the infectious feature of yawning.	

Questions 24–26

Do the following statements agree with the information given in Reading Passage 2? Write **TRUE**, **FALSE** *or* **NOT GIVEN**.

24 Some participants in Platek's first experiment did not understand why they were asked to look at people yawning.

25 The results from the second experiment indicate the link between yawning and empathy.

26 Yawning shows a positive impact on the recovery from brain damage brought by a stroke.

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Reading Passage 3

Incorrect Terminology in Science

A Science plays an increasingly significant role in people's lives, making the faithful communication of scientific developments more important than ever. Yet such communication is fraught with challenges that can easily distort discussions, leading to unnecessary confusion and misunderstandings.

B Some problems stem from the esoteric nature of current research and the associated difficulty of finding sufficiently faithful terminology. Abstraction and complexity are not signs that a given scientific direction is wrong, as some commentators have suggested, but are instead a tribute to the success of human ingenuity in meeting the increasingly complex challenges that nature presents. They can, however, make communication more difficult. But many of the biggest challenges for science reporting arise because in areas of evolving research, scientists themselves often only partly understand the full implications of any particular advance or development. Since that dynamic applies to most of the scientific developments that directly affect people's lives: global warming, cancer research, diet studies — learning how to overcome it is critical to spurring a more informed scientific debate among the broader public.

C Ambiguous word choices are the source of some misunderstandings. Scientists often employ colloquial terminology, which they then assign a specific meaning that is impossible to fathom without proper training. The term "relativity," for example, is intrinsically misleading. Many interpret the theory to mean that everything is relative and there are no absolutes. Yet although the measurements any observer makes depend on his coordinates and reference frame, the physical phenomena he measures have an invariant description that transcends that observer's particular coordinates. Einstein's theory of relativity is really about finding an invariant description of physical phenomena. True, Einstein agreed with the idea that his theory would have been better named "Invarianten theory." But the term — "relativity" — was already entrenched at the time for him to change.

D "The uncertainty principle" is another frequently abused term. It is sometimes interpreted as a limitation on observers and their ability to make measurements. But it is not about intrinsic limitations on any one particular measurement; it is about the inability to precisely measure particular pairs of quantities simultaneously. The first interpretation is perhaps more engaging from a philosophical or political perspective. It's just not what science is about.

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E Even the word "theory" can be a problem. Unlike most people, who use the word to describe a passing conjecture that they often regard as suspect, physicists have very specific ideas in mind when they talk about theories. For physicists, theories entail a definite physical framework embodied in a set of fundamental assumptions about the world that lead to a specific set of equations and predictions — ones that are borne out by successful predictions. Theories aren't necessarily shown to be correct or complete immediately. Even Einstein took the better part of a decade to develop the correct version of his theory of general relativity. But eventually both the ideas and the measurements settle down and theories are either proven correct, abandoned or absorbed into other, more encompassing theories.

F "Global warming" is another example of problematic terminology. Climatologists predict more drastic fluctuations in temperature and rainfall — not necessarily that every place will be warmer. The name sometimes subverts the debate, since it lets people argue that their winter was worse, so how could there be global warming? Clearly "global climate change" would have been a better name. But not all problems stem solely from poor word choices. Some stem from the intrinsically complex nature of much of modern science. Science sometimes transcends this limitation: remarkably, chemists were able to detail the precise chemical processes involved in the destruction of the ozone layer, making the evidence that chlorofluorocarbon gases (Freon, for example) were destroying the ozone layer indisputably.

G A better understanding of the mathematical significance of results and less insistence on a simple story would help to clarify many scientific discussions. For several months, Harvard was tortured by empty debates over the relative intrinsic scientific abilities of men and women. One of the more amusing aspects of the discussion was that those who believed in the differences and those who didn't used the same evidence about gender-specific special ability. How could that be? The answer is that the data shows no substantial effects. Social factors might account for these tiny differences, which in any case have an unclear connection to scientific ability. Not much of a headline when phrased that way, is it? Each type of science has its own source of complexity and potential for miscommunication. Yet there are steps we can take to improve public understanding in all cases. The first would be to inculcate greater understanding and acceptance of indirect scientific evidence. The information from an unmanned space mission is no less legitimate than the information from one in which people are on board.

H This doesn't mean never questioning an interpretation, but it also doesn't mean equating indirect evidence with blind belief, as people sometimes suggest. Second, we might need different standards for evaluating science with urgent policy implications than research with purely theoretical value. When scientists say they are not certain about their predictions, it doesn't necessarily mean they've found nothing substantial. It would be

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better if scientists were more open about the mathematical significance of their results and if the public didn't treat math as quite so scary. Statistics and errors, which tell us the uncertainty in a measurement, give us the tools to evaluate new developments fairly.

I But most importantly, people have to recognize that science can be complex. If we accept only simple stories, the description will necessarily be distorted. When advances are subtle or complicated, scientists should be willing to go the extra distance to give proper explanations and be patient about the truth. Even so, some difficulties are unavoidable. Most developments reflect work in progress, so the story is complex because no one yet knows the big picture.

Questions 27–31

Choose the correct letter, **A**, **B**, **C** *or* **D**.

27 Why is the faithful communication of science important?

A Science plays an increasingly significant role in people's lives.
B Science is fraught with challenges the public are interested in.
C The nature of complexity in science communication leads to confusion.
D Scientific inventions are more important than ever before.

28 What is the reason for many of the biggest challenges in science reporting?

A Phenomena such as global warming, cancer research and diet studies are too complex.
B Scientists themselves often only partly understand the Theory of Evolution.
C Scientists do not totally comprehend the impact of certain scientific advancements.
D Scientists themselves often only partly understand the esoteric nature of communication.

29 According to the 3rd paragraph, the reference to the term and example of "theory of relativity" is to demonstrate

A the theory of relativity is about an invariant physical phenomenon.
B scientific terms can sometimes cause misunderstandings.
C the term "relativity" is designed to mislead the public.
D everything is relative and there is no absolute existence.

30 Which one is a good example of an area where scientists used appropriate terminology?

A The uncertainty principle
B Global warming
C The theory of relativity
D Freon's destructive process on the ozone layer

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31 What is a surprising finding of the Harvard debates in the passage?

A There are equal intrinsic scientific abilities of men and women.

B Both sides used the same data in support of their argument.

C The scientific data usually shows no substantial figures to support a debated idea.

D Social factors might have a clear connection to scientific ability.

Questions 32–35

Do the following statements agree with the claims of the writer in Reading Passage 3? Write **YES**, **NO** *or* **NOT GIVEN**.

32 "Global warming" scientifically refers to greater fluctuations in temperature and rainfall rather than a universal temperature rise.

33 More media coverage of "global warming" would help the public to recognize the phenomenon.

34 Harvard debates should focus more on female scientists than male scientists.

35 Indirect scientific evidence is always less reliable than direct scientific evidence.

Questions 36–40

Complete the summary of the paragraphs of Reading Passage 3. Write **NO MORE THAN TWO WORDS** *from the Reading Passage for each answer.*

Science communication is fraught with challenges that can easily distort discussions, leading to unnecessary confusion and misunderstandings. Firstly, ambiguous 36 _____ are the source of some misunderstandings. Common people without proper training do not understand clearly or deeply a specific scientific meaning via the 37 _____ that scientists often employ. For example, uncertainty is not about 38 _____ limitations on any particular 39 _____. What's more, even the word "theory" can be a problem. Theories aren't necessarily shown to be correct or complete immediately since scientists often develop better versions of specific theories — a good example is the theory of 40 _____. Thus, most importantly, people have to recognize that science can be complex.

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IELTS Academic Reading — Test 2

Volume 10 · 3 Passages · 40 Questions

Reading Passage 1

A Brief History of Glassmaking

The first known glass objects can be traced back to the ancient Eastern empires, in about 3500 BC; building on existing knowledge about glazes, a new material, including sand and metal oxides, was developed. This material — glass — was initially used for small pieces of jewellery and inlays in small statues. Then it was used increasingly for vessels made to hold perfumes, oils, and fragrances. Transparency was not a characteristic of these works. Of greater interest to the early glass artisans was the ability to reproduce the colour and structures of highly valued semi-precious stones.

By about 450 BC, artisans in the glass workshops of the Persian Empire had learned to apply other techniques to glassmaking, probably producing vessels of considerable size on a turning wheel. Cutting techniques were also used for some of their complex decorations. Meanwhile, the Greeks still concentrated their energies on ceramics, and used glass simply for coloured inlays in sculptures — especially for magnificent monumental statues in public sites.

However, in the 1st and 2nd centuries BC, an industry of luxury glass items of various forms did emerge in former Greek areas strongly influenced by eastern cultures. This industry was in turn to have a profound effect on the glass art of the early Roman Empire.

The rise of the Roman Empire (1st century BC) was accompanied by an increase in glass production in the Mediterranean region. Initially glass retained its status as a luxury material — indeed from the earliest years of its development it had been generally considered equal to precious stones and valuable metals. Coloured glass bowls, mosaic-glass plates, and glass vessels with gold décor were very popular with the wealthy. However, as the empire expanded, the steps from luxury to mass production, and from decorative objects to vessels for daily use, were gradually taken.

The grounding for this was the Syrian glassmakers' discovery at the beginning of the 1st century BC that glass, when melted, can be made into shapes by blowing down a hollow metal tube. Thus it became possible for these craftsmen to produce a large variety of vessels for different uses — from simple bottles to more decorative and complex pieces. Starting with the leading Syrian glassworks, glass as a manufactured product spread across

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the entire area of Roman rule. With mass production, glass increasingly became a trade item — of no minor importance in convincing the so-called 'barbarians' from the border areas such as Germany of the superiority of Roman civilization.

The well-planned network of roads in the Roman Empire ensured that the new shapes of manufactured glass objects, and the newly developed techniques used to produce them, were quickly spread and adopted. Wherever suitable materials were found, the same or very similar glass forms, with hardly any variation in quality, were made. This is true of the Italian peninsula, including Rome itself, where numerous workshops producing high-quality goods existed, as well as of Spain and along the shores of the Black Sea. Because of their particularly well-suited sand and large deposits of soda ash, however, the main centres of glass production remained in Syria, Mesopotamia, and Lower Egypt.

Starting in the 1st century AD, small vessels and utensils in Roman households came to be made predominantly of wheel-blown glass. Beside the large number of everyday small vessels for storing fragrances and medicinal extracts, there were bowls in all shapes — some formed on the turning-wheel, some blown — plates, lamps, and much more. Even vessels technically very difficult to produce have been found, such as playful bottles in the shape of birds or other animals.

Whereas colour characterised the appearance of early Roman glass — in keeping with the Greek tradition — from the 2nd century AD, the transparency and also lightness of the material as well as its shiny surface, came to be valued more. Usually it was left in its natural state; typical are vessels having a blue-green coloration of varying intensity. There was, though, an increasing tendency to apply intricate details, and a few coloured accents can be observed. Different coloured glass was also used for simple handles, or for one or two threads or strings of extra glass wrapped around the rims to give emphasis to the vessels' mouths.

In the late Roman period (3rd and 4th centuries AD) the tendency towards complexity was also reflected in the growing popularity of cut and engraved work. Blowing and the use of moulds were developed in new directions, surpassing the designs of the early period of glass-blowing. Later vessels such as the so-called shellfish beaker represented a climax in glassmaking of the later Roman period: fish and similar objects were blown separately, formed with special tools, and melted onto the walls of simply-shaped vessels.

Roman glass art continued to be exemplary, and much of the world-famous Venetian glass art from the early 15th century was based on the revival of such methods as thread-glass decoration or mosaic glass. In fact, most methods of glassmaking and glass-decoration changed very little in the centuries following the Roman Empire. This is true of the development of European glass art, even up to the early 19th century. Until chemistry became an established scientific discipline in the second quarter of that century,

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glassmaking technology underwent only minor improvements. Even now, however, several of the techniques used in the Roman Empire remain a puzzle that modern glass technicians cannot resolve.

Questions 1–6

Do the following statements agree with the information given in Reading Passage 1? Write **TRUE**, **FALSE** *or* **NOT GIVEN**.

1 By about 450 BC, glass had become as important as ceramics in Greece.

2 The glassmakers of the early Roman Empire made copies of luxury Greek items.

3 With the rise of the Roman Empire, glass objects became more common.

4 Syrian glassmakers created many different types of objects by blowing glass.

5 The Roman road network provided a safer way of transporting glass than previous methods.

6 Roman glass vessels produced in the 1st century AD were restricted to simple designs.

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Questions 7–13

Complete the notes below. Write **NO MORE THAN TWO WORDS** *from the passage for each answer.*

DEVELOPMENTS IN GLASSMAKING — 2ND CENTURY AD

- greater emphasis on glass having a (7) ———, lightness and transparency
- complex (8) ——— were often added, as well as coloured accents
- extra threads of glass used to highlight the (9) ——— of vessels

LATE ROMAN PERIOD

- high point of later Roman glassmaking can be seen in containers like the (10) ———
- certain tools were used to create shapes, which were then attached to simple vessels

LATER DEVELOPMENTS

- from the 15th century, Venetians used old techniques like (11) ——— and thread-glass decoration
- acceptance of (12) ——— as a subject for serious study in the 19th century led to improvements in glassmaking technology
- some methods dating back to the Roman Empire continue to be a (13) ——— for today's glassmakers

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Reading Passage 2

The Return of the Black-Footed Ferret

The rise, fall, and rise of one of North America's rarest mammals

A The black-footed ferret (Mustela nigripes) is a small carnivorous mammal, similar in length to a domestic cat, and is the only ferret species native to North America. For at least 100,000 years they lived in an area that extended almost uninterrupted from southern Canada to northern Mexico. But numbers fell dramatically throughout the 19th and 20th centuries, to such an extent that the black-footed ferret was declared 'possibly extinct'. Then, almost miraculously, it returned from the dead.

B Within the genus Mustela, the ferret belongs to the subgenus Putorius, of which there are only three extant species: the European polecat; the Siberian polecat; and the black-footed ferret. The European polecat lives in open forests and meadows, and is thought to be the ancestor of the domestic ferret. The Siberian polecat looks more similar to the black-footed ferret, and leads a similar life on open grasslands and semidesert regions across Russia, China and Siberia.

C Ferrets probably evolved in Europe between three and four million years ago, and dispersed into North America across an ice-age land bridge at least two million years ago. They then advanced south-eastward to the Great Plains through ice-free passageways. The black-footed ferret's history in North America is inextricably linked to the creature which became almost its sole prey — the prairie dog. These small rodents are a type of ground squirrel which live in networks of underground tunnels, known as burrows. The black-footed ferret's short legs and long, slim body make it superbly adapted for life in the burrow systems of its prey.

D Spending virtually all their time underground, black-footed ferrets have always been elusive. Ferrets are never mentioned in accounts of the early explorers or pioneers who crossed the Great Plains by wagon train during the early to mid-1800s. They were, however, occasionally listed in records kept by fur companies in the upper Missouri River basin during that period. Black-footed ferrets were not officially recognized by scientists until 1851, when a description and drawing of one featured in a book by the renowned naturalist John James Audubon and Reverend John Bachman. For over two decades, no other specimens were seen, and scientists even began to question their existence. Then, in 1874, a Dr Elliott Coues issued a request for specimens through the popular magazine The American Sportsman, and was soon rewarded. With these he was able to add to Audubon's description, and confirm the existence of the black-footed ferret.

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E In the decades that followed, European settlement across North America changed the landscape dramatically, with vast areas being transformed into crop and grazing land. As agriculture and ranching expanded, the black-footed ferret's main prey, the prairie dog, became a target for extermination, and was eventually eradicated from 98% of its original range. The result was devastating for black-footed ferret populations throughout the continent. Even today, the prairie dog is still trapped and poisoned, because ranchers believe that it competes with livestock for grass — a notion that belies recent research showing that prairie dogs and cattle feed on different plants.

F As late as the 1950s, black-footed ferrets were still thought to occur in low densities throughout most of their historic range. However, by the 1960s, the only known population was a small colony in south-western South Dakota. In 1967, the US Fish and Wildlife Service classified the black-footed ferret as an endangered species, and in 1974 the South Dakota colony disappeared. Finally, in January 1979, the last ferret in a captive-breeding programme died — making the black-footed ferret the first mammal to be declared extinct in the Northern Hemisphere in the 20th century.

G But that was not the end of the story. In 1981, a farm dog called Shep carried an odd-looking creature to its owner's ranch house in Meeteetse, Wyoming. The rancher put the carcass in his fridge and alerted state officials. The dog had found a black-footed ferret — which led to efforts again being made to save the species. An intensive search in the region led to the discovery of a 130-strong population. This wild colony thrived for a while, but disaster struck again and the number of animals was drastically reduced by the disease distemper — probably contracted from domestic dogs. In 1985, in a final bid to save the species, the remaining 18 individuals in the colony were trapped for a new captive-breeding programme. This project was successful, and there are now more than 2,000 black-footed ferrets in captivity across the US and Canada.

H A reintroduction programme began in 1991. Before they can be reintroduced into their natural habitat, captive ferrets have to learn to hunt and escape from predators. They are first held in pens on the prairie, where they are fed prairie dog meat. Once the ferrets associate these animals with food, keepers unleash live prey into the pens, and they soon realise that they must hunt or go hungry. Then, to get the ferrets used to living underground, they have to be tempted into the holes with food.

I Reintroduction programmes are now underway across the central USA, but there is one location that is key. The Janos area on the US-Mexico border is the obvious place to reactivate the continent's wild black-footed ferret population, being one of the most important remaining grassland ecosystems in the Americas, and home to an abundance of prairie dogs. In the last report, the future of the black-footed ferret finally looks promising.

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Questions 14–16

Complete the sentences below. Write **NO MORE THAN TWO WORDS** *from the passage for each answer.*

The black-footed ferret shares most characteristics with its relative, the (14) _____

North American ferrets are thought to have originally come from (15) _____

The (16) _____ is the black-footed ferret's primary source of food throughout their time in North America.

Questions 17–22

Look at the following events and the list of dates below. Match each event with the correct date.

LIST OF DATES

A 1851
B 1874
C 1967
D 1974
E 1979
F 1981
G 1985
H 1991

Q	Statement	Answer
17	An appeal for evidence of black-footed ferrets was published.	
18	A scheme for releasing captive-born black-footed ferrets into the wild was launched.	
19	The black-footed ferret first appeared in a scientific work.	
20	The discovery of a single black-footed ferret led to renewed hope for the species.	
21	North America's last wild-born black-footed ferrets were taken into captivity.	
22	Efforts to save the black-footed ferret were officially announced as having failed.	

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Questions 23–26

Reading Passage 2 has nine paragraphs, **A–I**. *Which paragraph contains the following information?*

Q	Statement	Answer
23	reasons for the sharp decrease in black-footed ferret populations	
24	the procedure by which black-footed ferrets are taught to survive in the wild	
25	mention of doubts about whether there was such an animal as a black-footed ferret	
26	reference to a site which is ideal for the release of black-footed ferrets into the wild	

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Reading Passage 3

Rights for Apes

Humans have rights, other animals don't — no matter how human-like they are. Kate Douglas investigates the controversial campaign to get that changed.

A growing number of people across the world are attempting to persuade governments to grant great apes rudimentary 'human rights'. They argue that great apes are enough like us to deserve special treatment over other animals. The campaign has gathered momentum over the past few years and is stirring up a good deal of controversy. Various religious groups and the human rights charity, Amnesty International, point out that we have a long way to go to secure rights for humans before we start thinking about animals. Other critics argue that the proposal is not radical enough and should be extended to other intelligent social animals such as elephants and dolphins, and potentially to all species.

The Great Ape Project (GAP) is an international organisation that has been lobbying for legal rights for apes since 1993. The project has had some degree of success. New Zealand considered the possibility of extending human rights to great apes in 1999 as part of its new animal welfare bill. In the end, the bill did not go as far as granting apes individual legal rights, but did give them special status. Testing or teaching involving apes now requires government approval and must demonstrate that any likely benefits are not outweighed by harm to the individual animal. In effect that means no biomedical testing, only studies that increase our understanding of these species. The practice of using apes in biomedical research has also effectively ended in Europe now. So with all this legal protection in place, what's the big deal about individual rights? GAP campaigners say that these regulations don't go far enough and they still allow people who own great apes to lawfully neglect their needs. Michele Stumpe, a lawyer based in the US, points out that without rights, animals are mere property and their owners have no obligation to consider their best interests. Legally, however, only 'persons' can have 'rights', which is why the law needs to be changed to recognise the 'personhood' of apes.

Stumpe says that one reason for the resistance is the use of terms like 'rights' and 'personhood'. 'It seems like a radical idea and that scares people, but when we talk about rights, it really just means protection in layman's terms,' she says. For the philosopher Peter Singer (Princeton University), the opposition runs deeper than just semantics. He says the great apes are 'the victims of arbitrary discrimination', or what he calls 'speciesism'. He makes an argument on moral grounds for extending human rights to all beings that show intelligence and awareness — including some level of self-awareness — and have emotional and social needs. That means going beyond the boundaries of our own species — and therein lies the problem. 'It's that gulf between humans and animals that people want to maintain,' he says. Others take a similar view. It requires a radical shift in

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people's ideas about themselves, according to Ian Redmond, chief consultant for the UN Great Apes Survival Project. Gary Francione, a leading animal-rights lawyer from Rutgers University in Newark, New Jersey, and a member of the original GAP group, calls it 'spiritual superiority'. 'We just think we're special because we're human.'

Those less sympathetic to the animal rights agenda see things rather differently. Some argue that beyond the species barrier lies a slippery slope. 'Mice share around 90 per cent of human DNA: should they get 90 per cent of human rights?' asks geneticist Steve Jones from University College, London. He is concerned that giving rights to great apes would be the beginning of the end of all research with animals. Singer accepts that there is a question about where you draw the line, but sees no reason for that to stop society from taking the first step. 'The strongest and clearest case is with great apes, but we're open to arguments that it could be extended to other mammals,' he says.

Some take it even further. Francione no longer supports the idea of extending rights to great apes on the basis that their minds are like ours. Instead, he argues that all sentient beings should have just one right: the right not to be treated as the property of humans. He believes his approach has the advantage of simplicity, because sentience — the ability to feel pain or distress — is an objective quality, and because it would bring an end to captive animals altogether. It would make it illegal to breed any animals, including agricultural animals and even pets. His ideas highlight a logical inconsistency at the heart of the GAP agenda: it seeks to give legal rights to animals but does not give them the right to be free. Francione's approach is certainly several steps too far for many people.

It is not clear what GAP's proposals mean for wild apes, and even some of the animals' most outspoken defenders are not convinced that giving them human rights is the best way to protect them. Primatologist Frans de Waal, from the Yerkes Regional Primate Research Center in Atlanta, US, says 'The concept of rights applies only to those capable of carrying responsibilities within our society.' He believes the emphasis should be on the obligations humans have towards other animals, both for their care and for their conservation.

Redmond, on the other hand, suggests that applying GAP principles to wild populations might lead to increased penalties for killing apes or even give them land rights, though he admits the latter is unlikely. At the very least, an increased regard for the well-being of apes among people in western countries would lead to a decreased demand for animals being kidnapped from the wild and then exported to zoos, wildlife parks or private collectors.

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Questions 27–31

Complete the summary using the list of words below.

LIST OF WORDS

A experience **B** well-being **C** knowledge **D** regulations **E** charities **F** research **G** dolphins **H** needs **J** species

Some people believe that the campaign to grant human rights to the great apes does not go far enough. They want it to include other (27) _____ that are judged to demonstrate understanding. One country introduced (28) _____ which went some way to protecting the apes. As a result, anyone who is interested in using apes for (29) _____ must first obtain the official go-ahead from the state. They will have to prove that any advances gained do not come at the expense of the animal's (30) _____. So researchers can now only work on projects that may lead to the acquisition of further (31) _____ about the way apes function and interact.

Questions 32–36

Choose the correct answer.

(32) Why does Stumpe refer to 'layman's terms'?

A to show that many people hold extreme views about animals
B to suggest that campaigners are reluctant to criticise the law
C to suggest that such inflated terminology is unnecessary
D to argue that most people are not interested in protecting animals

(33) According to Singer, what is the root of the problem?

A Levels of intelligence and awareness differ between species.
B People disagree on how intelligent animals like apes are.
C Moral principles are not easy for everyone to understand.
D Human beings want to see themselves as separate from animals.

(34) If Francione's current idea was implemented,

A sentient animals could not be kept in cages.
B any form of large-scale agriculture would cease to exist.
C great apes would be granted all the same rights as humans.
D people could no longer offer shelter to an animal.

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35 What is the writer's attitude to Francione's theory?

A She is suspicious of the reasoning behind it.

B She is doubtful of its chances of being generally accepted.

C She is reassured by the impartial way it is expressed.

D She is pleased by its logical development.

36 How does de Waal change the perspective of the animal rights discussion?

A He recommends focusing on the way people look after animals.

B He would give rights only to those animals with a moral sense.

C He says that society should learn to be responsible for all of humanity.

D He thinks that conservation of endangered species should come first.

Questions 37–40

Do the following statements agree with the claims of the writer in Reading Passage 3? Write **YES**, **NO** *or* **NOT GIVEN**.

37 Particularly valuable animals in captivity are more likely to be protected by law.

38 de Waal highlights the need to give human rights to higher order animals.

39 If GAP beliefs were implemented in the wild, the survival rate of the larger apes would increase.

40 If people in Western countries, for example, learned to consider apes' needs, fewer apes would be captured for sale.

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IELTS Academic Reading — Test 3

Volume 10 · 3 Passages · 40 Questions

Reading Passage 1

Sleep Study on Modern-Day Hunter-Gatherers Dispels Popular Notions

A The sleep troubles common in modern life have long been blamed on our industrial society, from the city lights, long work hours and commutes, to caffeine and the Internet. Sleep researchers often look back on a time when humans were able to get more rest by sleeping and waking to the rhythms of the sun. It turns out that this may not be quite right. In fact, our ancestors may not have been getting the recommended eight hours of sleep, either.

B In a recent study, researchers traveled all over the world to examine sleep in some of the world's last remaining hunter-gatherer societies — the Hadza of Tanzania, the San of Namibia, and the Tsimane of Bolivia. Cut off from media, electricity and other distractions, these pre-industrial societies are thought to sleep the way humans did more than 10,000 years ago. Traveling to where they lived, often in humid, remote locations, researchers used medical devices to record the sleeping habits of 94 of these tribespeople and ended up collecting data representing 1,165 days.

C They found very similar sleep patterns despite their geographic isolation. On average, all three groups sleep a little less than 6.5 hours a night, do not take naps, and don't go to sleep when it gets dark. Like many of us, the Hadza, San, and Tsimane spend more time in bed — from 6.9 to 8.5 hours — than they do actually sleeping. This adds up to a sleep efficiency that is very similar to today's industrial populations.

D According to Jerome Siegel, director of the University of California's Center for Sleep Research, evidence suggests sleep habits may not be environmental or cultural, but central to the physical makeup of humans. These findings question the millions of dollars that have been spent on research that tries to explain why some sleepers get only about six hours of sleep a night. Also, such findings question whether lack of sleep is a cause of obesity, mood disorders, and other physical and mental illnesses which have become so common in recent decades. Scientists have documented that people's energy often falls in the mid-afternoon. Some have suggested that it's because we've managed to suppress a natural desire for a nap.

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E However, the new study provides evidence that this is unlikely, and that napping was actually rare in hunter-gatherer societies. The researchers estimated that naps may have occurred on up to 7 percent of winter days and 22 percent of summer days. They noted that their devices were only good at detecting longer naps, so it is possible that some of the study subjects took naps that were short, perhaps 15 minutes or less.

F Another fascinating finding from the study had to do with the circadian rhythms, our daily activity cycles related to sunlight. Instead of going to sleep right at dusk, tribespeople were staying awake an average of between 2.5 and 4.4 hours after sunset. All three tribes had fires going, but the light itself was much lower than you might get from a light bulb. They did, however, have a tendency to wake up anywhere between an hour before and an hour after sunrise.

G Siegel and his co-authors investigated this further by looking into the significance of temperature. They found that it also played a big role, though it was somewhat less important than light in influencing sleep patterns. They wrote that “sleep in both the winter and summer usually occurred during the period of cooling and that waking times usually occurred near the height of the daily warming trend.”

H The tribespeople that were studied are different from people living in modern conditions in a number of respects. Importantly, almost none of them were troubled by sleeplessness. In interviews with the researchers conducted through interpreters, only 1.5 to 2.5 percent of the study subjects said they had severe difficulties sleeping more than once a year. This figure is far lower than the 10 to 30 percent recorded in many industrialized countries today. Siegel suggested that “mimicking aspects of the natural environment” may therefore help treat some sleep disorders.

I The tribespeople are also much healthier. Not a single one is overweight, indicating their overall higher levels of physical fitness. They also tended to have healthier hearts. Thus comes a critical question. If we can't blame our health problems on our lack of sleep, could it be that the reason we feel so unrested is because of poor health?

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Questions 1–4

Do the following statements agree with the information given in Reading Passage 1? Write **TRUE**, **FALSE** *or* **NOT GIVEN**.

1 Scientists studied hunter-gatherer societies because their sleep patterns are assumed to be similar to those of humans more than 10,000 years ago.

2 The medical devices used in the research were specially designed for humid conditions.

3 Researchers found that tribespeople stayed longer in bed than inhabitants of industrialised regions.

4 Jerome Siegel believes that environment and culture have little effect on sleep patterns.

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Questions 5–13

Complete the notes below. Write **ONE WORD ONLY** *from the passage for each answer.*

NEW EVIDENCE ON SLEEP PATTERNS — NEW IDEAS ABOUT NAPPING

- scientists have recorded an afternoon drop in (5) _____
- studies of hunter-gatherer societies show that napping was rare and occurred more often during the (6) _____
- the devices may not have detected (7) _____ naps

DAILY ACTIVITY CYCLES

- the tribespeople went to sleep several hours after sunset with (8) _____
- there was little light
- tribespeople usually woke up around (9) _____
- scientists found that (10) _____ had almost as much influence on sleep patterns as light

DIFFERENCES BETWEEN TRIBESPEOPLE AND PEOPLE IN INDUSTRIALISED REGIONS

- (11) _____ is something that very few of the tribespeople suffered from
- the environment of tribespeople may have been more suitable for sleep
- tribespeople had better fitness than industrialised populations and were not (12) _____
- tribespeople also had stronger (13) _____

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Reading Passage 2

Bristlecone Pines

A On the dry, windswept mountain tops of the Great Basin in the western United States, at altitudes of over 3,000 m, grow the bristlecone pines (Pinus longaeva, Pinus aristata). The bristlecone has adjusted to places on earth that no other tree wants to inhabit, and in these harsh environments has flourished. Bristlecones don't grow very tall, 18.3 m at the most, but usually much less, while the girth of the largest one has been measured at 11.2 m. What makes bristlecone pines truly remarkable is their longevity: their average age is over 1,000 years, with the oldest specimen — an individual known as Methuselah — dated at 4,723 years. This makes it not only the world's oldest living tree; it is also the earth's oldest living inhabitant. The extreme age of bristlecone pines was not known about until the 1950s, and was only discovered then thanks to the efforts of Edmund Schulman. In 1932, Schulman had begun his career in dendrochronology as an assistant to A. E. Douglass of the Laboratory of Tree-ring Research at the University of Arizona.

B Douglass had noted that the wide rings of certain species of trees were produced during wet years and, inversely, narrow rings during dry seasons. The more a tree's rate of growth has been affected by such environmental factors, the more variation in ring-to-ring growth will be present. This variation is referred to as sensitivity and the lack of ring variability is called complacency. Patterns of ring variability are important for establishing chronologies, which allow precise dating of climatic conditions in the past. Samples taken from trees of unknown age can be studied for matches with samples from trees of a known age with known sequences of growth. Using this process, when the rings match or are found to have overlapping patterns, the chronology can be extended further back in time.

C In his search for older trees that would provide more extensive chronologies, Schulman learned that certain species of trees in the upper-forest zones, growing under stressful conditions, showed sensitive records of drought in their growth-ring sequences. The short, distorted, and dwarfed trees of the upper tree lines were now his focus. During 1954 and 1955, an extensive survey of bristlecone stands from California to Colorado was carried out by Schulman and his assistant, C. W. Ferguson. They found the oldest trees at elevations of 3,048 to 3,354 m, often in seemingly impossible locations, on dolomite soils where no other plant life could survive. These trees showed large areas of deadwood and thin strips of living bark. The trees growing in the most extreme conditions, with scant soil and moisture, seemed to be the oldest. This prompted Schulman to write: 'The capacity of these trees to live so fantastically long may, when we come to understand it fully, perhaps serve as a guidepost on the road to the understanding of longevity in general.'

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D So what factors do account for the longevity of bristlecone pines? Spring comes to the bristlecone pines in early May with the melting of snow and rising temperatures. In this subalpine zone there are only three warm summer months, often only six weeks, to produce growth and reserves for overwintering. All of this must be accomplished on a mere 25.4 cm of rainfall. To live so long under such conditions, the bristlecone has established several strategies. Firstly, bristlecone needles can live twenty to thirty years; thus, adding new foliage to that already on the tree takes little energy. The long-lived needles provide a stable photosynthetic capacity to sustain the tree over years of severe stress. Another strategy for surviving is the gradual dieback of living bark when the tree is damaged because of fire, lightning, drought, or storms. This reduction of tissue that the crown of the tree has to supply with nutrients balances the effect of any damage sustained. The surviving parts remain quite healthy. As an example, Pine Alpha, at over 4,000 years old, is over a metre in diameter, yet has only a 25 cm strip of living bark to support it. Finally, invasions from bacteria, fungi, or insects that prey upon most plants are unknown to the bristlecones due to their highly resinous wood.

E Environmental factors also play their part. The dry air common in the subalpine region helps preserve the trees from rotting. The oldest bristlecones live in the most exposed sites, with considerable space between each tree. The longevity of the bristlecone needles and the inability of other plants to grow in the dolomite soil preferred by the bristlecones make for little leaf litter. This distance in between, combined with the lack of ground cover, ensures that if a tree sustains a lightning strike and catches fire, the fire does not spread to surrounding trees. Thus, the bristlecone pines have survived for unknown centuries. The current threat is from all the people who come to visit them. As a result, Methuselah, the oldest tree, is not marked or identified with a plaque due to the threat of vandalism. The recording of past events provided by these trees, along with their great beauty, is too valuable for us to lose.

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Questions 14–17

Reading Passage 2 has five paragraphs, **A–E**. *Which paragraph contains the following information? NB You may use any letter more than once.*

Q	Statement	Answer
14	an explanation of differences in ring patterns	
15	mention of a human danger to bristlecone pines	
16	an account of research conducted on bristlecone pines	
17	an advantage of the bristlecone pines' habitat	

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Questions 18–20

Choose the correct letter, **A**, **B**, **C** *or* **D**.

18 According to the writer, the most interesting fact about bristlecone pines is their

A habitat.
B height.
C width.
D age.

19 A pattern of narrow rings is caused by

A alternating rainy and dry seasons.
B a lack of rain.
C restricted growth in some trees.
D high altitude.

20 Schulman chose to study trees growing at high levels because they were

A shorter than similar trees.
B growing in unusual shapes.
C much older than other trees.
D the only surviving plants.

Questions 21–26

Complete the notes below. Write **NO MORE THAN TWO WORDS** *from the passage for each answer.*

FACTORS IN THE SURVIVAL OF BRISTLECONE PINES — A. FEATURES OF BRISTLECONE PINES

- making new needles requires 21 _____
- trees damaged by weather or fire can exist with only small areas of 22 _____
- bristlecone wood is very 23 _____

B. FEATURES OF THE ENVIRONMENT

- trees don't decay because of the 24 _____
- fires don't spread easily because of the 25 _____ between trees and absence of 26 _____

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Reading Passage 3

Loss of Rare Languages

"Obviously we must do some serious rethinking of our priorities, lest linguistics go down in history as the only science that presided obviously over the disappearance of 90 percent of the very field to which it is dedicated." — Michael Krauss, The World's Languages in Crisis

A Ten years ago, Michael Krauss sent a shudder through the discipline of linguistics with his prediction that half the 6,000 or so languages spoken in the world would cease to be uttered within a century. Unless scientists and community leaders directed a worldwide effort to stabilize the decline of local languages, he warned, nine tenths of the linguistic diversity of humankind would probably be doomed to extinction. Krauss's prediction was little more than an educated guess, but other respected linguists had been sounding similar alarms. Keneth L. Hale of the Massachusetts Institute of Technology noted in the same journal issue that eight languages on which he had done fieldwork had since passed into extinction. A 1990 survey in Australia found that 70 of the 90 surviving Aboriginal languages were no longer used regularly by all age groups. The same was true for all but 20 of the 175 Native American languages spoken or remembered in the US, Krauss told a congressional panel in 1992.

B Many experts in the field mourn the loss of rare languages, for several reasons. To start, there is scientific self-interest; some of the most basic questions in linguistics have to do with the limits of human speech, which are far from fully explored. Many researchers would like to know which structural elements of grammar and vocabulary — if any — are truly universal and probably therefore hardwired into the human brain. Other scientists try to reconstruct ancient migration patterns by comparing borrowed words that appear in otherwise unrelated languages. In each of these cases, the wider the portfolio of languages you study, the more likely you are to get the right answers.

C Despite the near constant buzz in linguistics about endangered languages over the past 10 years, the field has accomplished depressingly little. 'You would think that there would be some organized response to this dire situation, some attempt to determine which languages can be saved and which should be documented before they disappear,' says Sarah G. Thomason, a linguist at the University of Michigan at Ann Arbor. 'But there isn't any such effort organized in the profession. It is only recently that it has become fashionable enough to work on endangered languages.' 'Six years ago,' recalls Douglas H. Whalen of Yale University, 'when I asked linguists who was raising money to deal with these problems, I mostly got blank stares.' So, Whalen and a few other linguists founded the Endangered Languages Fund. In the five years to 2001 they were able to collect only $80,000 for research grants. A similar foundation in England, directed by Nicholas Ostler, has raised just $8,000 since 1995.

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D But there are encouraging signs that the field has turned a corner. The Volkswagen Foundation, a German charity, just issued its second round of grants totaling more than $2 million. It has created a multimedia archive at the Max Planck Institute for Psycholinguistics in the Netherlands that can house recordings, grammars, dictionaries and other data on endangered languages. To fill the archive, the foundation has dispatched field linguists to document Aweti (100 or so speakers in Brazil), Ega (about 300 speakers in Ivory Coast), Waima'a (a few hundred speakers in East Timor), and a dozen or so other languages unlikely to survive the century. The Ford Foundation has also edged into the arena. Its contributions helped to reinvigorate a master-apprentice program created in 1992 by Leanne Hinton of Berkeley and Native Americans worried about the imminent demise of about 50 indigenous languages in California. Fluent speakers receive $3,000 to teach a younger relative (who is also paid) their native tongue through 360 hours of shared activities, spread over six months. So far about 5 teams have completed the program, Hinton says, transmitting at least some knowledge of 25 languages. 'It's too early to call this language revitalization,' Hinton admits. 'In California the death rate of elderly speakers will always be greater than the recruitment rate of young speakers. But at least we prolong the survival of the language.' That will give linguists more time to record these tongues before they vanish.

E But the master-apprentice approach hasn't caught on outside the U.S., and Hinton's effort is a drop in the sea. At least 440 languages have been reduced to a mere handful of elders, according to the Ethnologue, a catalogue of languages produced by the Dallas-based group SIL International that comes closest to global coverage. For the vast majority of these languages, there is little or no record of their grammar, vocabulary, pronunciation or use in daily life. Even if a language has been fully documented, all that remains once it vanishes from active use is a fossil skeleton, a scattering of features that the scientist was lucky and astute enough to capture. Linguists may be able to sketch an outline of the forgotten language and fix its place on the evolutionary tree, but little more. 'How did people start conversations and talk to babies? How did husbands and wives converse?' Hinton asks. 'Those are the first things you want to learn when you want to revitalize the language.'

F But there is as yet no discipline of 'conservation linguistics,' as there is for biology. Almost every strategy tried so far has succeeded in some places but failed in others, and there seems to be no way to predict with certainty what will work where. Twenty years ago in New Zealand, Maori speakers set up 'language nests,' in which preschoolers were immersed in the native language. Additional Maori-only classes were added as the children progressed through elementary and secondary school. A similar approach was tried in

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Hawaii, with some success — the number of native speakers has stabilized at 1,000 or so, reports Joseph E. Grimes of SIL International, who is working on Oahu. Students can now get instruction in Hawaiian all the way through university.

G One factor that always seems to occur in the demise of a language is that the speakers begin to have collective doubts about the usefulness of language loyalty. Once they start regarding their own language as inferior to the majority language, people stop using it for all situations. Kids pick up on the attitude and prefer the dominant language. In many cases, people don't notice until they suddenly realize that their kids never speak the language, even at home. This is how Cornish, and some dialects of Scottish Gaelic, died out, and it is how Irish Gaelic is still only rarely used for daily home life in Ireland, 80 years after the republic was founded with Irish as its first official language.

H Linguists agree that ultimately, the answer to the problem of language extinction is multilingualism. Even uneducated people can learn several languages, as long as they start as children. Indeed, most people in the world speak more than one tongue, and in places such as Cameroon (279 languages), Papua New Guinea (823) and India (387) it is common to speak three or four distinct languages and a dialect or two as well. Most Americans and Canadians, to the west of Quebec, have a gut reaction that anyone speaking another language in front of them is committing an immoral act. You get the same reaction in Australia and Russia. It is no coincidence that these are the areas where languages are disappearing the fastest. The first step in saving dying languages is to persuade the world's majorities to allow the minorities among them to speak with their own voices.

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Questions 27–33

Reading Passage 3 has eight paragraphs, **A–H**. *Choose the correct heading for paragraphs from the list of headings below.*

Example: Paragraph C — **vi** *(given)*

LIST OF HEADINGS

i Inadequate data
ii Consensus on an initial recommendation for saving dying out languages
iii Positive gains in protection efforts
iv Minimum requirement for saving a language
v Potential threat to minority languages
vi A period when there was an absence of real efforts made *(used in example)*
vii Uncertainty about the success of native language programs
viii Lack of confidence in young speakers as a negative factor
ix Practice in several developing countries
x Value of minority languages to linguists
xi Government participation in language field

Q	Paragraph	Answer
27	Paragraph A	
28	Paragraph B	
29	Paragraph D	
30	Paragraph E	
31	Paragraph F	
32	Paragraph G	
33	Paragraph H	

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Questions 34–38

Use the information in the passage to match the people with opinions or deeds below. Choose the appropriate letters.

LIST OF PEOPLE

A Nicholas Ostler
B Michael Krauss
C Joseph E. Grimes
D Sarah G. Thomason
E Keneth L. Hale
F Douglas H. Whalen

Q	Statement	Answer
34	Reported about the success of a language conservation practice in Hawaii	
35	Predicted that many languages would disappear soon	
36	Experienced loss of languages he had worked on personally	
37	Raised language funds in England	
38	Not enough effort on saving languages until recent work	

Questions 39 and 40

Choose the correct letter, **A**, **B**, **C** *or* **D**.

39 What is the real result of the master-apprentice program sponsored by the Ford Foundation?

A Teach children how to speak native languages
B Revive some endangered languages in California
C Postpone the dying date for some endangered languages
D Increase communication between students

40 What should majority language speakers do according to the last paragraph?

A They should teach their children endangered languages
B They should learn at least four languages
C They should show their loyalty to a dying language
D They should be more tolerant of minority language speakers

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PART TWO

Answer Keys

Complete answer keys for Tests 1–3 of Volume 10. Check your answers only after finishing each test.

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Answer Key

Reading Test 1 (Volume 10)

Passage 1 — Questions 1–13

1. E

PASSAGE EVIDENCE	Lyell and Hooker "pressed Darwin to present his conclusions so that he would not lose priority"

2. A

PASSAGE EVIDENCE	Suess theorized "its upper crust shrinks and wrinkles like the withering skin of an aging apple"

3. D

PASSAGE EVIDENCE	Wegener believed in continental drift while dismissing the "landbridge" theory

4. B

PASSAGE EVIDENCE	Wallace published "the Geographical Distribution of Animals in 1876"

5. C

PASSAGE EVIDENCE	"Darwin and his young contemporary Alfred Russel Wallace reached the conclusion that life had evolved by natural selection"

6. B

PASSAGE EVIDENCE	"the fittest would survive"

7. F

PASSAGE EVIDENCE	"a distinct faunal boundary, called 'Wallace's line,' between an Asian realm... and an Australian realm"

8. G

PASSAGE EVIDENCE	"animals long established in southwestern Australia had an affinity with animals in South Africa"

9. migrated

PASSAGE EVIDENCE	"plants and animals had once migrated across now-submerged intercontinental landbridges"

10. withering skin

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PASSAGE EVIDENCE	"shrinks and wrinkles like the withering skin of an aging apple"

11. tectonic plates

PASSAGE EVIDENCE	"shifting, colliding and separating tectonic plates"

12–13. dispersalism, vicarianism

PASSAGE EVIDENCE	"The discussion now concerns 'dispersalism' versus 'vicarianism'"

Passage 2 — Questions 14–26

14. interval

PASSAGE EVIDENCE	"they come in bouts with an interval of about 68 seconds"

15. complex distinction

PASSAGE EVIDENCE	"men are significantly less likely to cover their mouths, which may indicate complex distinction in genders"

16. breathing

PASSAGE EVIDENCE	"coordinated within the hypothalamus of the brain, the area that also controls breathing"

17. stretching

PASSAGE EVIDENCE	"a link between yawning and stretching as early as the end of the first prenatal trimester"

18. brain damage

PASSAGE EVIDENCE	"paralyzed on one side of their body because of brain damage caused by a stroke"

19. E

PASSAGE EVIDENCE	"the basic yawn lasts about six seconds and they come in bouts with an interval"

20. B

PASSAGE EVIDENCE	"Yawning is an ancient, primitive act... One species of penguins yawns as part of mating"

21. F

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PASSAGE EVIDENCE	"we don't always stretch when we yawn"

22. D

PASSAGE EVIDENCE	"it's also a good indicator if you're empathizing with me and paying attention"

23. H

PASSAGE EVIDENCE	"contagious yawning, by contrast, implicates many higher brain regions"

24. TRUE

PASSAGE EVIDENCE	"We literally had people saying, 'Why am I looking at people yawning?'"

25. TRUE

PASSAGE EVIDENCE	"the part of the brain which reacted was the part scientists believe controls empathy"

26. NOT GIVEN

PASSAGE EVIDENCE	"It is not known whether the associated response is a positive prognosis for recovery" — explicitly unconfirmed either way

Passage 3 — Questions 27–40

27. A

PASSAGE EVIDENCE	"Science plays an increasingly significant role in people's lives, making the faithful communication of scientific developments more important than ever"

28. C

PASSAGE EVIDENCE	"scientists themselves often only partly understand the full implications of any particular advance or development"

29. B

PASSAGE EVIDENCE	the "relativity" example illustrates that "ambiguous word choices are the source of some misunderstandings"

30. D

PASSAGE EVIDENCE	"chemists were able to detail the precise chemical processes involved in the destruction of the ozone layer... indisputably"

31. B

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PASSAGE EVIDENCE	"those who believed in the differences and those who didn't used the same evidence"

32. YES

PASSAGE EVIDENCE	"Climatologists predict more drastic fluctuations in temperature and rainfall — not necessarily that every place will be warmer"

33. NOT GIVEN

PASSAGE EVIDENCE	the passage suggests renaming to "global climate change," not more media coverage — not addressed

34. NOT GIVEN

PASSAGE EVIDENCE	no recommendation about which sex the debate should focus on is given

35. NO

PASSAGE EVIDENCE	"The information from an unmanned space mission is no less legitimate than the information from one in which people are on board"

36. word choices

PASSAGE EVIDENCE	"Ambiguous word choices are the source of some misunderstandings"

37. colloquial terminology

PASSAGE EVIDENCE	"Scientists often employ colloquial terminology"

38–39. intrinsic; measurement

PASSAGE EVIDENCE	"it is not about intrinsic limitations on any one particular measurement"

40. general relativity

PASSAGE EVIDENCE	"Even Einstein took the better part of a decade to develop the correct version of his theory of general relativity"

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Answer Key

Reading Test 2 (Volume 10)

Passage 1 — Questions 1–13

1. FALSE

PASSAGE EVIDENCE	"the Greeks still concentrated their energies on ceramics, and used glass simply for coloured inlays"

2. NOT GIVEN

PASSAGE EVIDENCE	no mention of Romans copying Greek luxury items specifically

3. TRUE

PASSAGE EVIDENCE	"the steps from luxury to mass production... were gradually taken"

4. TRUE

PASSAGE EVIDENCE	"it became possible for these craftsmen to produce a large variety of vessels for different uses"

5. NOT GIVEN

PASSAGE EVIDENCE	roads ensured techniques "were quickly spread and adopted" — no comparison to safety of previous methods

6. FALSE

PASSAGE EVIDENCE	"there were bowls in all shapes... plates, lamps, and much more... vessels technically very difficult to produce"

7. shiny surface

PASSAGE EVIDENCE	"the transparency and also lightness of the material as well as its shiny surface, came to be valued more"

8. details

PASSAGE EVIDENCE	"an increasing tendency to apply intricate details"

9. mouths

PASSAGE EVIDENCE	"threads or strings of extra glass wrapped around the rims to give emphasis to the vessels' mouths"

10. shellfish beaker

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PASSAGE EVIDENCE	"the so-called shellfish beaker represented a climax in glassmaking of the later Roman period"

11. mosaic glass

PASSAGE EVIDENCE	"based on the revival of such methods as thread-glass decoration or mosaic glass"

12. chemistry

PASSAGE EVIDENCE	"Until chemistry became an established scientific discipline in the second quarter of that century"

13. puzzle

PASSAGE EVIDENCE	"several of the techniques used in the Roman Empire remain a puzzle that modern glass technicians cannot resolve"

Passage 2 — Questions 14–26

14. Siberian polecat

PASSAGE EVIDENCE	"The Siberian polecat looks more similar to the black-footed ferret"

15. Europe

PASSAGE EVIDENCE	"Ferrets probably evolved in Europe between three and four million years ago"

16. prairie dog

PASSAGE EVIDENCE	"the creature which became almost its sole prey — the prairie dog"

17. B

PASSAGE EVIDENCE	"in 1874, a Dr Elliott Coues issued a request for specimens"

18. H

PASSAGE EVIDENCE	"A reintroduction programme began in 1991"

19. A

PASSAGE EVIDENCE	"not officially recognized by scientists until 1851, when a description and drawing of one featured in a book"

20. F

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PASSAGE EVIDENCE	"In 1981, a farm dog called Shep carried an odd-looking creature to its owner's ranch house"

21. **G**

PASSAGE EVIDENCE	"In 1985... the remaining 18 individuals in the colony were trapped for a new captive-breeding programme"

22. **E**

PASSAGE EVIDENCE	"in January 1979, the last ferret in a captive-breeding programme died"

23. **E**

PASSAGE EVIDENCE	"the black-footed ferret's main prey, the prairie dog, became a target for extermination"

24. **H**

PASSAGE EVIDENCE	"they are first held in pens on the prairie, where they are fed prairie dog meat"

25. **D**

PASSAGE EVIDENCE	"scientists even began to question their existence"

26. **I**

PASSAGE EVIDENCE	"The Janos area on the US-Mexico border is the obvious place to reactivate... population"

Passage 3 — Questions 27–40

27. **J — species**

PASSAGE EVIDENCE	"extended to other intelligent social animals such as elephants and dolphins, and potentially to all species"

28. **D — regulations**

PASSAGE EVIDENCE	"Testing or teaching involving apes now requires government approval"

29. **F — research**

PASSAGE EVIDENCE	"only studies that increase our understanding of these species"

30. **B — well-being**

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PASSAGE EVIDENCE	"any likely benefits are not outweighed by harm to the individual animal"

31. C — knowledge

PASSAGE EVIDENCE	"only studies that increase our understanding of these species"

32. C

PASSAGE EVIDENCE	"it really just means protection in layman's terms"

33. D

PASSAGE EVIDENCE	"It's that gulf between humans and animals that people want to maintain"

34. A

PASSAGE EVIDENCE	"all sentient beings should have just one right: the right not to be treated as the property of humans"

35. B

PASSAGE EVIDENCE	"Francione's approach is certainly several steps too far for many people"

36. A

PASSAGE EVIDENCE	"the emphasis should be on the obligations humans have towards other animals, both for their care and for their conservation"

37. NOT GIVEN

PASSAGE EVIDENCE	no mention of value affecting legal protection level

38. NO

PASSAGE EVIDENCE	de Waal says rights apply "only to those capable of carrying responsibilities" — he focuses on obligations, not rights

39. NOT GIVEN

PASSAGE EVIDENCE	Redmond only speculates about penalties/land rights, no survival-rate claim

40. YES

PASSAGE EVIDENCE	"an increased regard for the well-being of apes... would lead to a decreased demand for animals being kidnapped from the wild"

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Answer Key

Reading Test 3 (Volume 10)

Passage 1 — Questions 1–13

1. TRUE

PASSAGE EVIDENCE	"these pre-industrial societies are thought to sleep the way humans did more than 10,000 years ago"

2. NOT GIVEN

PASSAGE EVIDENCE	devices were used in humid locations, but no claim they were specially designed for that

3. NOT GIVEN

PASSAGE EVIDENCE	tribespeople's time in bed is compared to their own sleep, not directly to industrialised populations' time in bed

4. NOT GIVEN

PASSAGE EVIDENCE	"evidence suggests sleep habits *may not be* environmental or cultural" — hedged, not a definite belief either way

5. energy

PASSAGE EVIDENCE	"people's energy often falls in the mid-afternoon"

6. summer

PASSAGE EVIDENCE	"naps may have occurred on up to 7 percent of winter days and 22 percent of summer days"

7. short

PASSAGE EVIDENCE	"study subjects took naps that were short, perhaps 15 minutes or less"

8. fires

PASSAGE EVIDENCE	"All three tribes had fires going, but the light itself was much lower"

9. sunrise

PASSAGE EVIDENCE	"wake up anywhere between an hour before and an hour after sunrise"

10. temperature

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PASSAGE EVIDENCE	"looking into the significance of temperature. They found that it also played a big role"

11. **sleeplessness**

PASSAGE EVIDENCE	"almost none of them were troubled by sleeplessness"

12. **overweight**

PASSAGE EVIDENCE	"Not a single one is overweight"

13. **hearts**

PASSAGE EVIDENCE	"they also tended to have healthier hearts"

Passage 2 — Questions 14–26

14. **B**

PASSAGE EVIDENCE	"the wide rings of certain species of trees were produced during wet years"

15. **E**

PASSAGE EVIDENCE	"the threat of vandalism"

16. **C**

PASSAGE EVIDENCE	"an extensive survey of bristlecone stands from California to Colorado was carried out"

17. **E**

PASSAGE EVIDENCE	"The dry air common in the subalpine region helps preserve the trees from rotting"

18. **D**

PASSAGE EVIDENCE	"What makes bristlecone pines truly remarkable is their longevity"

19. **B**

PASSAGE EVIDENCE	"narrow rings during dry seasons"

20. **C**

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PASSAGE EVIDENCE	"They found the oldest trees at elevations of 3,048 to 3,354 m"

21. little energy

PASSAGE EVIDENCE	"adding new foliage to that already on the tree takes little energy"

22. living bark

PASSAGE EVIDENCE	"has only a 25 cm strip of living bark to support it"

23. resinous

PASSAGE EVIDENCE	"unknown to the bristlecones due to their highly resinous wood"

24. dry air

PASSAGE EVIDENCE	"The dry air common in the subalpine region helps preserve the trees from rotting"

25. distance

PASSAGE EVIDENCE	"considerable space between each tree"

26. ground cover

PASSAGE EVIDENCE	"combined with the lack of ground cover"

Passage 3 — Questions 27–40

27. v

PASSAGE EVIDENCE	Krauss's prediction that "nine tenths of the linguistic diversity of humankind would probably be doomed to extinction"

28. x

PASSAGE EVIDENCE	"the wider the portfolio of languages you study, the more likely you are to get the right answers"

29. iii

PASSAGE EVIDENCE	"there are encouraging signs that the field has turned a corner"

30. iv

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PASSAGE EVIDENCE	"all that remains once it vanishes from active use is a fossil skeleton" unless documentation happens first

31. **vii**

PASSAGE EVIDENCE	"succeeded in some places but failed in others, and there seems to be no way to predict with certainty"

32. **viii**

PASSAGE EVIDENCE	"Kids pick up on the attitude and prefer the dominant language"

33. **ii**

PASSAGE EVIDENCE	"the answer to the problem of language extinction is multilingualism"

34. **C**

PASSAGE EVIDENCE	"reports Joseph E. Grimes of SIL International, who is working on Oahu"

35. **B**

PASSAGE EVIDENCE	"Michael Krauss sent a shudder through the discipline of linguistics with his prediction"

36. **E**

PASSAGE EVIDENCE	"eight languages on which he had done fieldwork had since passed into extinction"

37. **A**

PASSAGE EVIDENCE	"A similar foundation in England, directed by Nicholas Ostler, has raised just $8,000"

38. **D**

PASSAGE EVIDENCE	"But there isn't any such effort organized in the profession. It is only recently that it has become fashionable"

39. **C**

PASSAGE EVIDENCE	"But at least we prolong the survival of the language"

40. **D**

PASSAGE EVIDENCE	"to persuade the world's majorities to allow the minorities among them to speak with their own voices"

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